

Roll No.:

END SEMESTER EXAMINATION DECEMBER 2025

Name of the Program: B.Tech

Semester: V

Name of the Course: Natural Language Processing and Computer Vision

Course Code: TCS 564

Time: 3 Hours

Maximum Marks: 100

Note:

- i. All questions are compulsory.
- ii. Answer any two sub questions among a, b & c in each main questions
- iii. Total marks in each main question are twenty.
- iv. Each questions carries 20 marks

Q1	(20Marks)	
(a)	How do you think emojis used by current generation impact tokenization? How will you handle acronyms like LOL, ASAP, ROFL in the NLP pipeline? What changes will you made in the NLP pipeline (tokenization, pos tagging etc.) to handle such emojis and acronyms.	CO1
(b)	Chunking in multi-language Hinglish text (mixing of Hindi and English) requires different approach. Propose how will you handle such multi language while performing chunking.	
(c)	Write a Python program that compares recall between stemming and no stemming in a simple retrieval task.	
Q2		
(a)	Define and Discuss Cosine Similarity. Find the cosine similarity between the following documents: Doc1: India is the most populated country in the world. Doc2: India has overtaken China as the most populated country in the world.	CO2
(b)	What are Word Embeddings? Differentiate between CBOW and Skip-gram using appropriate example.	
(c)	Using the example sentences below, compute the TF-IDF weight of each word in each document: Doc1: The cat sleeps on the mat. Doc2: A dog barks at the cat. Doc3: The mat is soft and warm.	
Q3		
(a)	What is Computer Vision and why do we use Computer Vision? Discuss some applications of Computer Vision. Images also have ambiguity. – Explain this statement very briefly.	CO3
(b)	What is a Color Model? What is Color Gamut? Discuss the RGB and HSB color model.	
(c)	Write complete code to perform sentiment analysis using NLTK or spaCy. Assume that you have a dataset of reviews of Restaurants in a TSV file. The dataset contains two columns, one contains the reviews and the other a value (1 or 0), 1 indicates positive review and 0 indicates negative review. There are total 1500 reviews. Perform all necessary preprocessing work. Make any other assumptions you require.	

Q4		
(a)	What is a Linear Shift Invariant System (LSIS)? Given that $f(x)$ convolves with $h(x)$ to produce $g(x)$, how can you obtain $h(x)$? Give a brief idea as to how the idea of Convolution is used in Image Processing.	CO4 & CO5
(b)	Differentiate between Sobel and Canny edge detection. Name an edge detection technique that uses double integration in place of single integration. Write code in OpenCV for Canny and Sobel Edge detection.	
(c)	What is basis of image segmentation by humans? Discuss briefly. Discuss any Clustering based method for segmentation using an example.	
Q5		
(a)	Explain the architecture of the CNN model that forms the backbone of YOLO. Briefly discuss any three improvements of current YOLO version.	CO5 & CO6
(b)	List some popular object detection framework. Write code in OpenCV for detecting a face in an image using Haar Cascades.	
(c)	Provide the working principles of the following cues: Relative Size, Texture Gradient, Interposition, Linear Perspective, Height in Plane, Light and Shadow, Binocular Disparity, Binocular Convergence	